

# PUSHKAR SINGH

Bangalore, India · Open to Visa Sponsorship

☎ +91-9410560129

✉ [singh.pushkar7570@gmail.com](mailto:singh.pushkar7570@gmail.com)

🌐 [linkedin.com/in/pushkar-singh1/](https://linkedin.com/in/pushkar-singh1/)

🐙 [github.com/pushkar7570/](https://github.com/pushkar7570/)

## Summary

AI/ML Engineer with 2+ years of experience building and deploying machine learning and generative AI solutions for enterprise workflows. Skilled in Python, data processing, model training, and integrating LLMs using RAG and LangGraph. Experienced in model evaluation, experimentation, and developing scalable AI systems with MLOps practices in production environments.

## Experience

### PwC - PricewaterhouseCoopers

Feb. 2024 – Present

#### Associate 2 (Gen AI)

##### • ACTA - Agentic Controls Testing Assistant

- Designed and implemented ACTA as a multi-agent LangGraph-based system, orchestrating parallel agents using RAG and semantic indexing to analyze multi-modal datasets, perform feature extraction, and automate audit workflows.
- Built and deployed on Azure using LangChain, Azure OpenAI, and Microsoft Graph connectors to extract, preprocess, and transform enterprise data, leveraging embeddings, OCR, and multimodal pipelines to generate structured insights and reduce manual effort by 60%.
- Implemented LLM-as-a-judge and cross-encoder based confidence scoring to evaluate model performance using metrics, analyse error patterns, and optimize model outputs, improving accuracy by 40% while optimizing latency through retrieval tuning.

##### • CLUTS - Custom GPT based solution

- Built CLUTS, a GPT-powered pipeline with multiple reasoning steps using prompt engineering and RAG to normalize controls into structured format, detect gaps, and standardize audit-ready outputs, and adapted it to client-specific requirements across 5+ projects within the practice.
- Deployed on Azure and developed with LangChain, Azure OpenAI, and RAG-based retrieval to generate test steps, sampling logic, and evidence requirements, enabling scalable control standardization and reducing manual audit effort.

### Insignia Consultancy Solutions

May 2023 – Aug. 2023

#### Microservices Developer Intern

- Collaborated with development team to design and implement Java SpringBoot microservices for subscription management, including input validation and transaction rollback.
- Developed REST endpoints, creating JPA entities and optimizing MySQL queries for accuracy and performance.

## Education

### Graphic Era Hill University, Dehradun

Sep. 2020 – Jun. 2024

B.Tech. in Computer Science & Engineering (8.25/10)

### Maria Assumpta Sr. Sec. Convent School, Kashipur

Mar. 2019 – Apr. 2020

Senior Secondary Education - XII (85%)

## Skills

- **AI Systems & tools:** Generative AI, Agentic AI, RAG, LangChain, LLMs, MCP Servers, Multi-Agent system
- Languages & Frameworks:** Python, Java, Fast API, System Design, SpringBoot Microservices, REST APIs
- Cloud, DevOps & Databases:** Azure OpenAI, Docker, MLOps, Git, DevOps, ChromaDB, MongoDB
- Professional Skills:** Problem Solving, Adaptability, Client Management, Structured Communication, Active Listening

## Projects

### Roxy - Interview Voice Assistant (Llama, FastAPI, RAG, LangGraph, ChromaDB, Whisper, Coqui)

- Designed architecture for Roxy using LangGraph as a multimodal interview system with multiple orchestrated agents handling - content analysis, questioning, evaluation & feedback.
- Built and deployed on cloud using FastAPI, ChromaDB, Whisper STT and Coqui TTS, leveraging RAG, embeddings to process PDFs, audio, and video, enabling voice-driven interviews with real-time evaluation and retrieval-backed responses.
- Implemented self-improving pipeline that refines prompts, retrieval, and evaluation using feedback-driven workflow analysis.

## Certifications & Publications

- Selected for **McKinsey Forward Program**, focused on problem solving, digital, and leadership skills.
- **AWS Certified Cloud Practitioner**
- **Published research** on breakout/reversal trading using candlestick-close and RSI divergence strategies. **IEEE Xplore**
- Awarded **PwC's Extra Miler Award & Quarterly LoS** for taking Agentic AI initiatives and project deliverables.